Ideation Phase
Brainstorm & Idea Prioritization Template

Date	31 January 2026
Team ID	LTVIP2026TMIDS80513
Project Name	TransLingua: AI-Powered Multi-Language Translator
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Brainstorm & idea prioritization

Before you collaborate

A Team gathering

- **Gopinadh Polukonda:** Technical feasibility and API integration.
- **Sk. Kulsum:** User scenarios and linguistic accuracy.
- **Karthikeya Seethepalli:** Session facilitation and idea prioritization.

B Set the goal

- **Scenario 1 (Business):** How can we ensure the AI maintains a professional "brand voice" instead of a literal word-for-word translation?
- **Scenario 2 (Academic):** How can we help researchers translate dense papers while keeping the original formatting and technical terminology intact?
- **Scenario 3 (Travel):** How can we make the Streamlit interface fast enough for a traveler standing in a train station with a spotty data connection?

C Learn how to use the facilitation tools

- As the facilitator, **Karthiekya** will set a strict 5-minute timer for each "How Might We" statement. When the buzzer sounds, the team must move to the next scenario (Business → Academic → Travel). This prevents the team from over-polishing one idea while ignoring others.
- **Kulsum** will ensure that ideas aren't just spoken but "seen." If someone suggests a "Business Dashboard," she will quickly sketch a rough layout or use icons to represent features. This helps the team align on what the actual UI in Streamlit will look like.
- **Gopinadh** will use his technical expertise to build momentum. Instead of saying "No, that's too hard to code," he will say, *"Yes, and we can achieve that by using the Gemini API's system instructions to preserve formatting."* This keeps the energy high and the ideas flowin

1 Define your problem statement

How might we use Generative AI to provide context-aware, professional-grade translations that preserve the specific tone and formatting required for business, academic, and travel use cases?

Step-2: Brainstorm, Idea Listing and Grouping

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Brainstorm

Gopinadh

- Implement "Context Tags" so users can tell the AI if the text is a legal contract or a casual chat.
- Create a "Latency Toggle" to switch between high-speed basic translation and deep-reasoning accurate translation
- Add an automated "Markdown/LaTeX Formatter" to keep academic papers looking professional.

Kulsum

- "Traveler's Quick-Key" interface with giant buttons for common phrases like "Where is the station?"
- A "Dual-Pane View" for researchers to see the original paper side-by-side with the translation.
- A "Brand Voice" upload feature where businesses can upload a glossary of terms they want the AI to always use.

Karthikeya

- An "AI Emotion Analyzer" that tells you the tone of the translated text
- Integration with a camera API to translate text from live images for travelers.
- A "History & Favorites" dashboard so users can save frequently used academic citations or business pitch phrases

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Group ideas

A. Contextual Intelligence (The "Pro" Mode)
Instead of a simple "translate" button, the group idea is to implement a **Context Selector**.

- **Business:** A toggle to enforce "Formal/Professional" tone and specific brand glossaries.
- **Academic:** A specialized mode that ignores LaTeX formulas and preserves citations while translating technical terminology.
- **Goal:** To move beyond literal word-for-word translation to "intent-based" translation.

B. Adaptive User Interfaces
The group suggests a **Scenario-Based UI** that changes based on the user's current need.

- **Travel Mode:** High-contrast, large-button interface optimized for mobile screens and quick voice-to-text input.
- **Research Mode:** A split-screen desktop view allowing for side-by-side comparison of source and target text.

C. Reliability & Feedback Loop
To ensure accuracy, the group proposes a **Tone & Accuracy Analyzer**.

- The AI provides a "Confidence Score" and a "Tone Summary" (e.g., "This translation sounds highly formal").
- A "back-translation" feature to verify the meaning by translating the result back into the source language.

Step-3: Idea Prioritization

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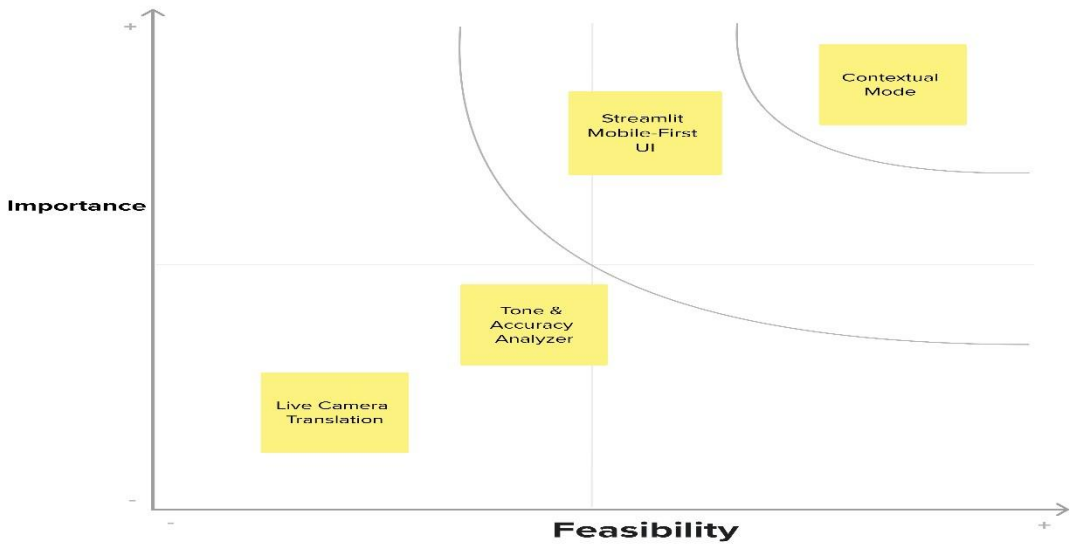
Prioritize